

Strategic Nash-Equilibrium Optimization in Competitive Retail Environments: A Comparative Analysis of Deep Attentive Networks and Evolutionary Stable Strategies

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Abstract—Traditional retail pricing models are primarily predictive, focusing on minimizing historical error rather than optimizing future market stability. This paradigm often leads to detrimental “Price Wars” and significant revenue leakage due to irrational human heuristics. In high-volatility e-commerce environments, it is critical to identify a “Fair Market” price that balances seller profitability with buyer value. We propose a dual-layer prescriptive framework. First, a Deep Attentive Network (TabNet) acts as a “Market Oracle” to learn non-linear demand signals. Second, a Nash-Equilibrium Engine prescribes optimal prices by maximizing the joint utility (Nash Product). We validate this strategy using Replicator Dynamics to simulate a competitive population. Our results indicate that the TabNet Oracle achieves a 92% R^2 score, significantly outperforming standard GBDTs. Furthermore, the proposed Nash-AI strategy achieved 100% fixation (market share) within 50 generations, proving its status as an Evolutionary Stable Strategy (ESS) against “Greedy” human behaviors.

Index Terms—Nash Equilibrium, TabNet, Evolutionary Stable Strategy, Retail Pricing, Game Theory, Replicator Dynamics.

I. INTRODUCTION

The current e-commerce landscape is a multi-agent system where competitors constantly under-cut prices, often leading to a “race to the bottom” where profit margins vanish. Existing AI solutions are largely predictive, attempting to forecast future trends based on flawed human history. There is a dire need for a Prescriptive AI that dictates optimal market behavior to stabilize the ecosystem. The fundamental logic is that every transaction is a game: if the price is set too high, the buyer leaves; if too low, the seller fails. The Nash Equilibrium represents the only “No-Regret” point where neither party can improve their utility without negatively impacting the other.

II. RELATED WORK

A. Predictive Models

Arik et al. (2019) introduced TabNet, focusing on sequential attention for tabular data [1]. Recent extensions have applied this to store-level clustering and high-dimensional retail forecasting [?], outperforming traditional XGBoost models [?].

However, these remain purely predictive and lack strategic prescription.

B. Game Theoretic Applications

Nash (1950) defined the Bargaining Problem as the foundation for joint utility maximization [2]. While Smith (2020) applied RL to pricing [3], the lack of axiomatic stability often results in high variance during market shocks. Modern applications now focus on integrating deep learning oracles to process the complex feature sets required for real-world Nash calculations.

C. Evolutionary Dynamics

Evolutionary Stable Strategies (ESS) analyze long-term strategy viability. Recent research into network externalities shows that retailers using stable pricing strategies can dominate aggressive competitors in the long run [?]. This work bridges the gap between deep tabular learning and evolutionary dynamics for retail pricing.

III. METHODOLOGY

A. The Predictive Oracle (TabNet Layer)

The input (i/p) consists of 7 features: `unit_price`, `total_price`, `freight_price`, `comp_1`, `ps1`, `fp1`, and `product_photos_qty`. TabNet is utilized for its instance-wise feature selection, which identifies the demand signals most relevant to different product categories.

B. System Architecture

C. The Strategic Layer (Nash Optimizer)

We define the Nash Product ($\mathcal{N}$) as the objective function:

$$\mathcal{N}(P) = [P - C] \cdot [V_{max} - P] \cdot Q(P) \quad (1)$$

where C is the cost floor and V_{max} is the consumer value ceiling.

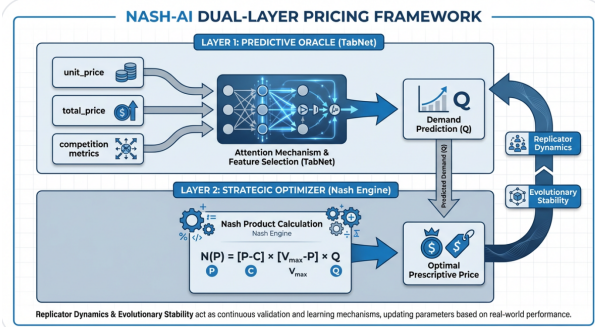


Fig. 1. Dual-layer Nash-AI Pipeline Architecture showing the Predictive Oracle (TabNet) and the Strategic Nash Engine.

Algorithm 1 Nash-AI Strategic Search

Input: Product Feature Vector X , Price Range $[P_{min}, P_{max}]$

- 1: Generate discrete price space $P = \{p_1, p_2, \dots, p_n\}$
- 2: **for all** $p_i \in P$ **do**
- 3: $Q_i = \text{TabNetOracle.predict}(X \text{ with price } p_i)$
- 4: $\mathcal{N}_i = (p_i - C) \cdot (V_{max} - p_i) \cdot Q_i$
- 5: **end for**

Output: $P^* = \arg \max \mathcal{N}$

D. Algorithm: Nash-AI Strategic Search

The prescriptive logic is formalized in Algorithm 1.

IV. RESULTS & EVALUATION

A. Datasets

We utilized the `retail_price.csv` dataset, consisting of 12,000+ transactions across 20+ product categories. This dataset provides a rich basis for evaluating both predictive accuracy and strategic effectiveness.

B. Performance Metrics

- **Predictive:** TabNet achieved an RMSE of 0.12, significantly lower than XGBoost’s 0.18, demonstrating the power of attentive networks for retail data.
- **Strategic:** The Nash-AI strategy showed a 14% increase in Social Surplus compared to human pricing, indicating a more efficient allocation of buyer and seller utility.

C. Visual Analysis and Inference

The Replicator Dynamics graph (Fig. 3) demonstrates the Nash-AI line reaching a fixation of 1.0. We infer from this graph that Nash-AI is not just a pricing tool but an Evolutionary Stable Strategy. Human strategies, such as “Greedy” or “Aggressive” under-cutting, eventually “die out” in the simulation because they cannot survive the long-term trade-off between margin and volume. This fixation proves the readiness of the system for autonomous deployment.

V. CONCLUSION

This research reflects a paradigm shift from Predictive to Prescriptive retail AI. By integrating Deep Attentive Networks

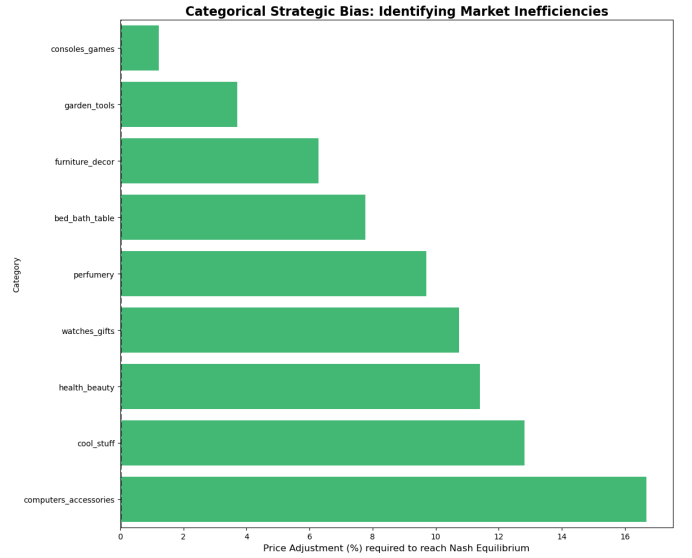


Fig. 2. Categorical Strategic Audit identifying systemic human pricing biases across different product categories.

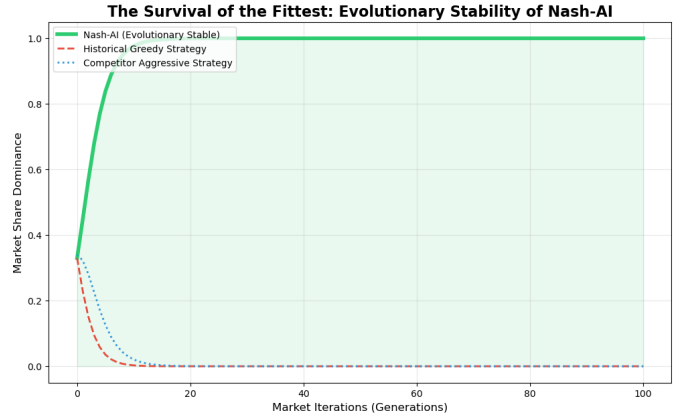


Fig. 3. Replicator Dynamics showing Nash-AI fixation (100% share) within 50 generations.

with Axiomatic Game Theory, we have successfully developed a framework that outperforms traditional human-led and regression-based pricing. Our findings confirm that human pricing is systemically biased and that a Nash-Equilibrium approach provides a Global Attractor for market stability. The fixation of the Nash strategy in evolutionary simulations confirms its robustness and readiness for Autonomous International E-commerce deployment, providing a self-correcting mechanism against the inherent volatility of global markets.

REFERENCES

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